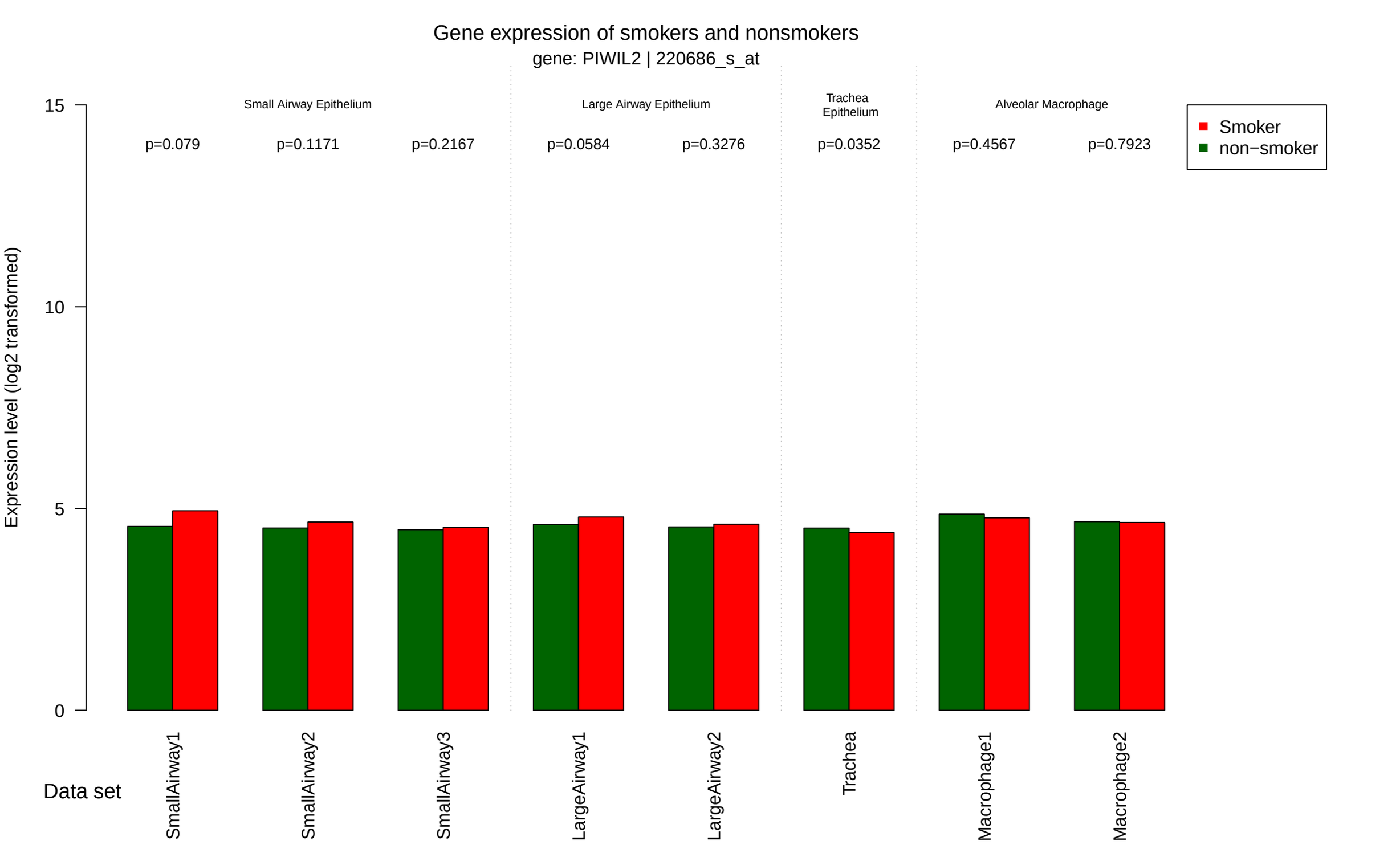
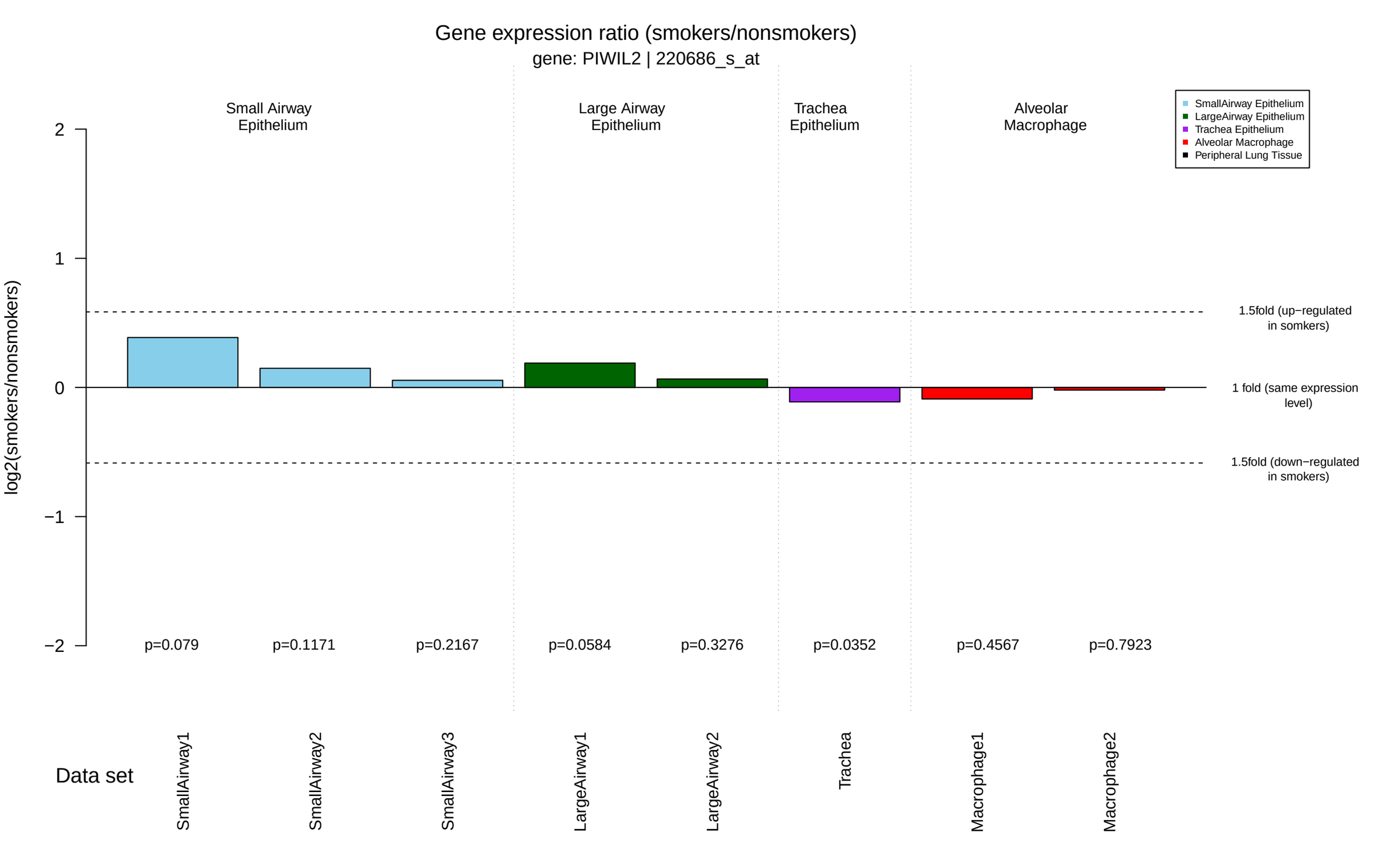
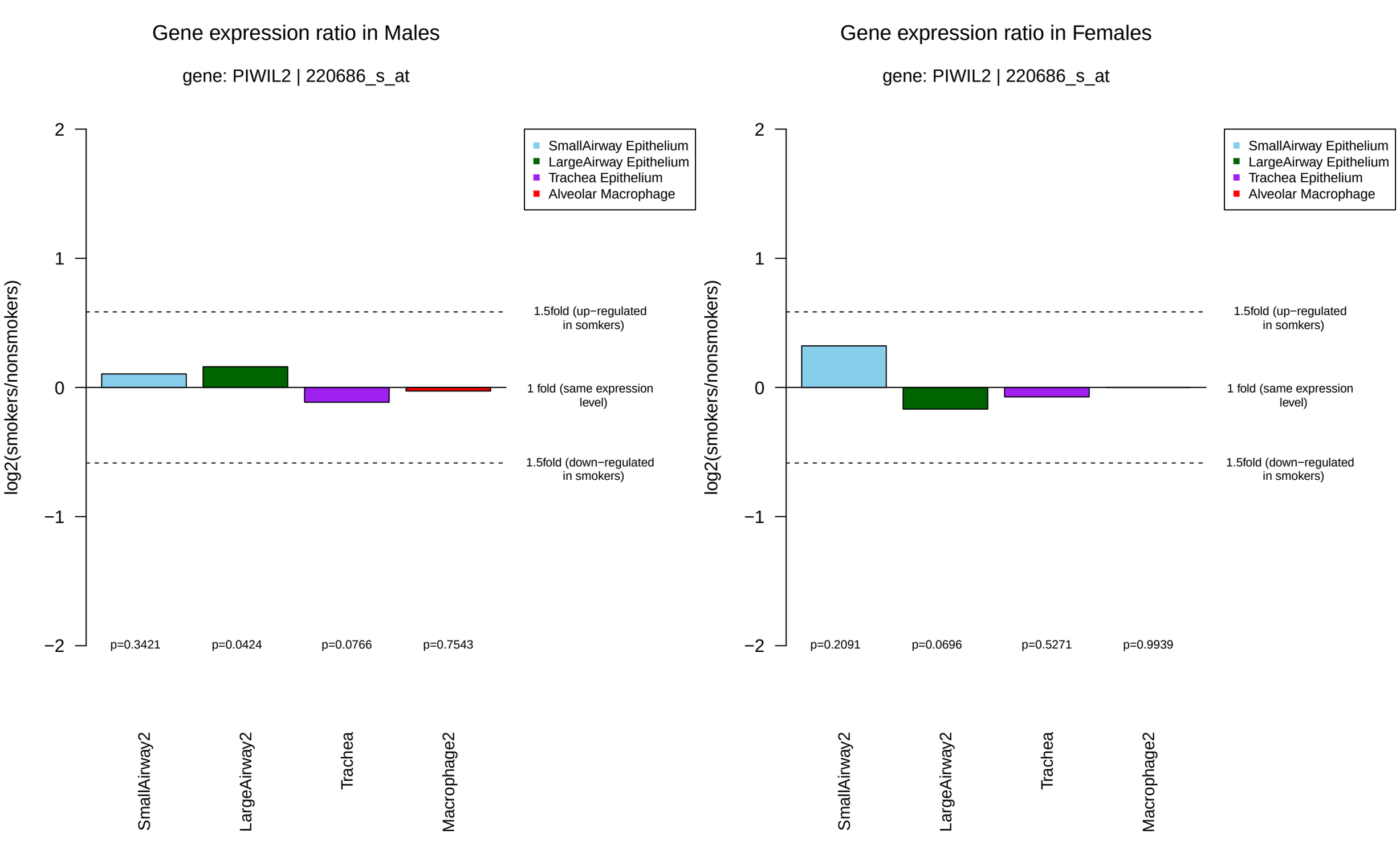
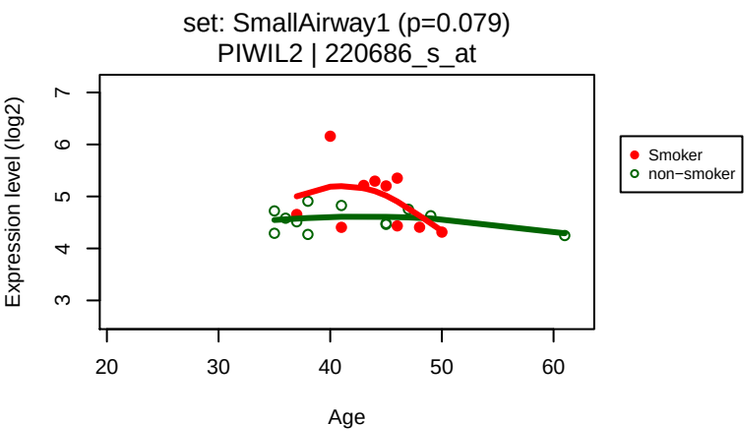
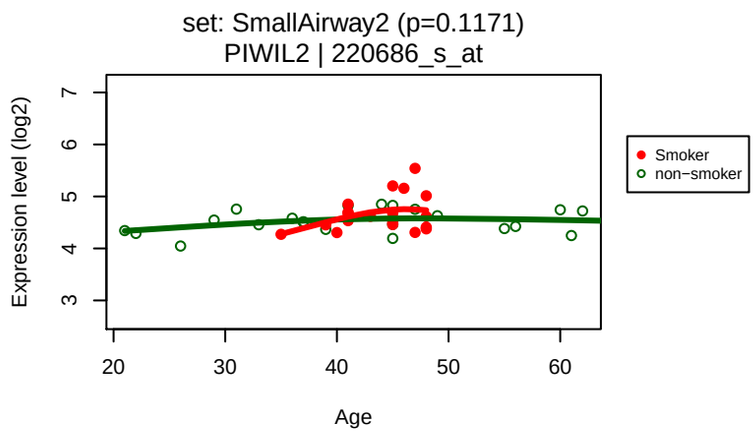
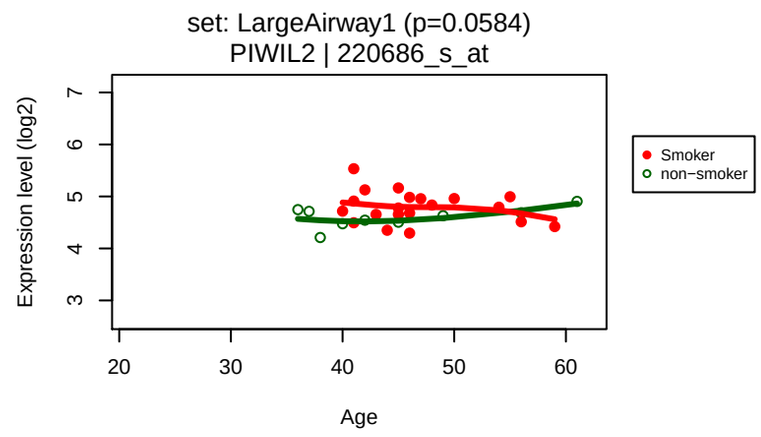

Gene expression of smokers and nonsmokers
gene: PIWIL2 | 220686_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.079
p=0.1171
p=0.2167
p=0.0584
p=0.3276
p=0.0352
p=0.4567
p=0.7923
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

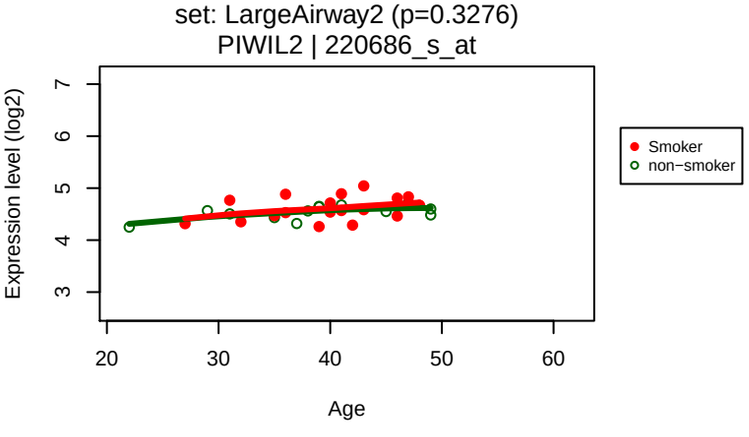

Gene expression ratio (smokers/nonsmokers)
gene: PIWIL2 | 220686_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.079
p=0.1171
p=0.2167
p=0.0584
p=0.3276
p=0.0352
p=0.4567
p=0.7923
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

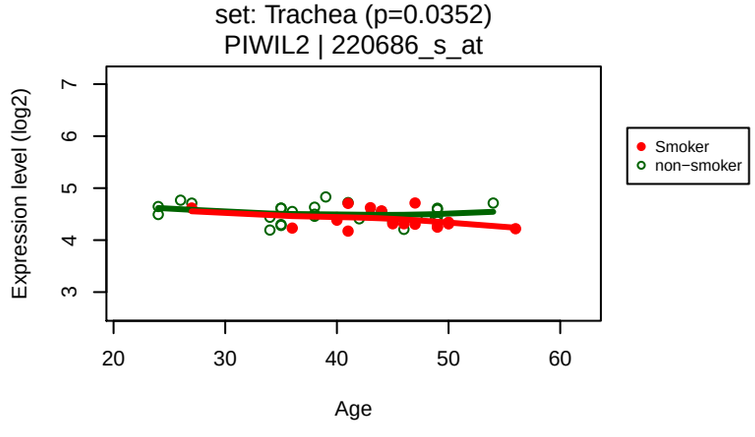

Gene expression ratio in Males
gene: PIWIL2 | 220686_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3421
p=0.0424
p=0.0766
p=0.7543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIWIL2 | 220686_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2091
p=0.0696
p=0.5271
p=0.9939
SmallAirway2
LargeAirway2
Trachea
Macrophage2

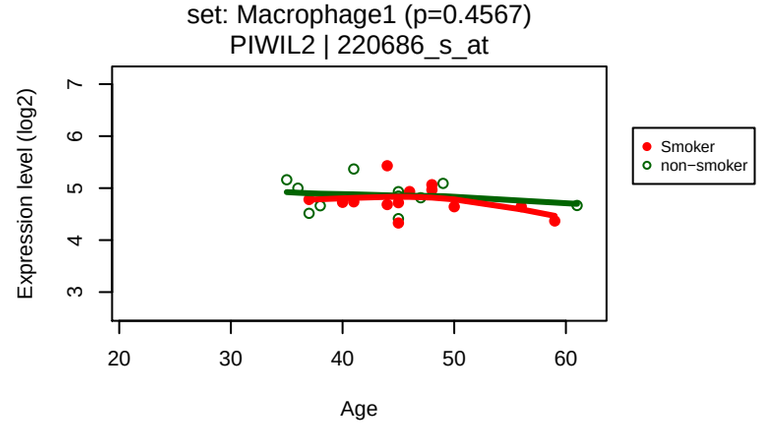

set: SmallAirway1 (p=0.079)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker

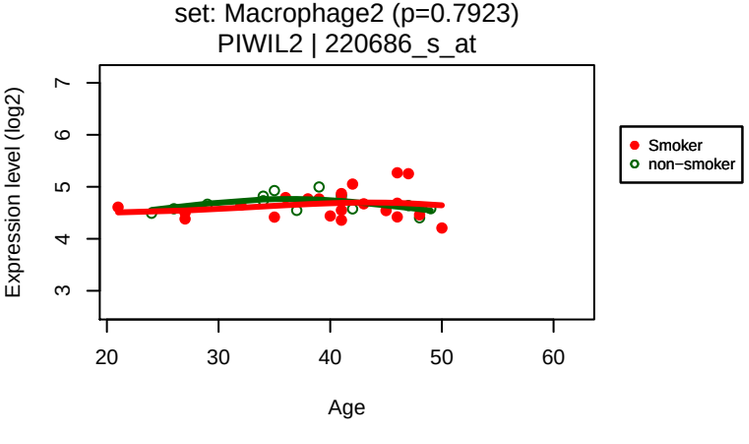

set: SmallAirway2 (p=0.1171)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker

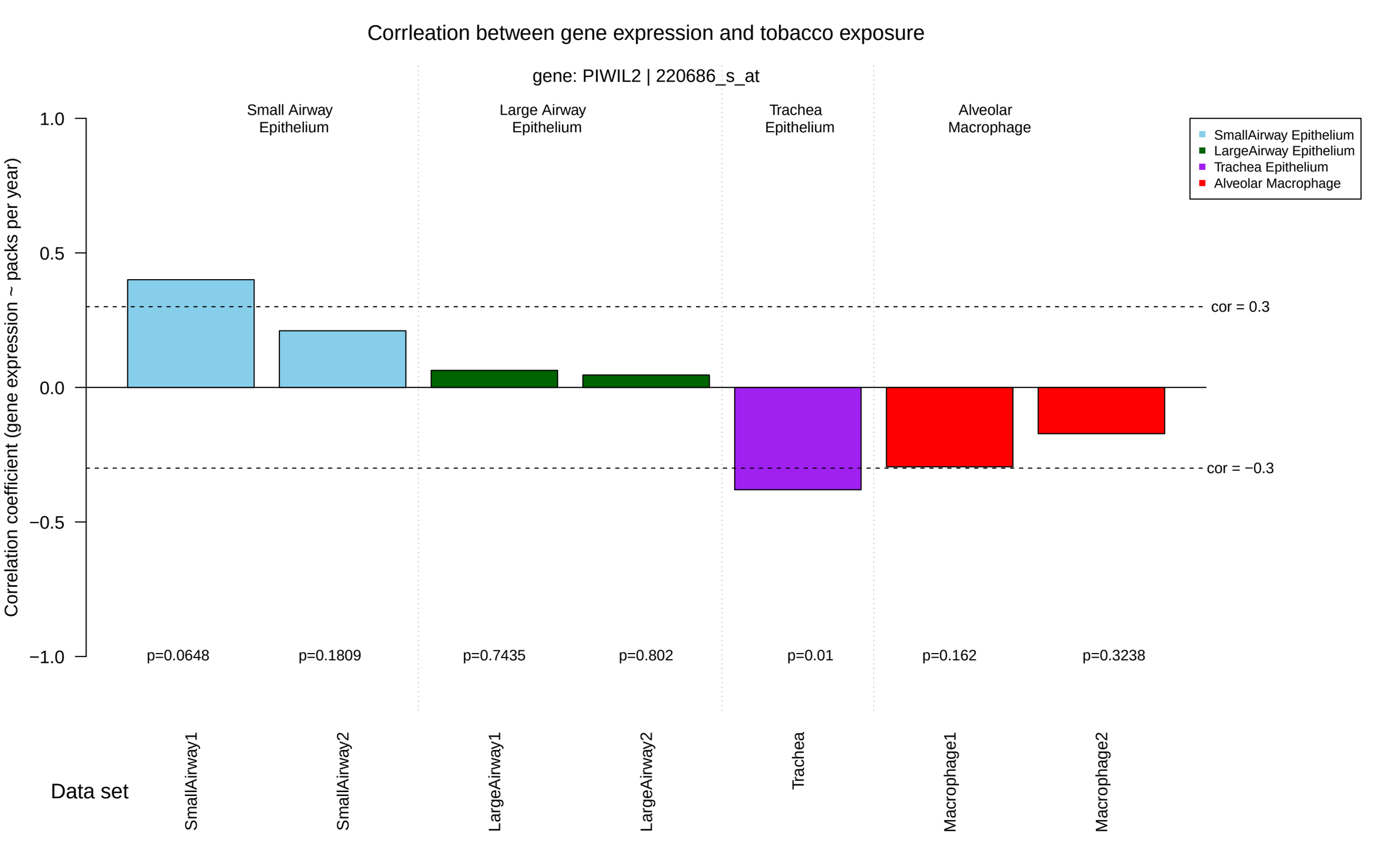

set: LargeAirway1 (p=0.0584)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3276)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0352)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4567)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7923)
PIWIL2 | 220686_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIWIL2 | 220686_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0648
p=0.1809
p=0.7435
p=0.802
p=0.01
p=0.162
p=0.3238
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2